Subspace

$$R = \underset{M\times K}{A}PA^H + \sigma^2 I$$

ULA $\longrightarrow$ A Vandermonde $\quad [e^{j\pi m\cos\theta_n}]_{M\times N}$ $\quad \boxed{\begin{matrix}1\\ e\\ e^2\\ e^{M-1}\end{matrix}}$

$\begin{cases} P \text{ diagonal} \\ \text{ULA} \end{cases} \longrightarrow R$ Toeplitz

Pf: $[R]_{m_1,m_2} = \sum_{n=1}^{N} P_n e^{j\pi(m_1-m_2)\cos\theta_n}$

ESPRIT 条件：ULA

Eigenvalue decomposition：

$R = \underset{M\times M}{U}\Lambda U^H$ 找最大 k 个 λ_n 作信号子空间

$= \underset{M\times k}{U_s}\underset{k}{\Lambda_s}U_s^H + \underset{(M-k)}{U_n}\Lambda_n U_n^H$

取 $U_{s1} := U_s$ 前 $M-1$ 行 $\quad U_{s2} := U_s$ 后 $M-1$ 行

求线性变换 $\Psi = \arg\min \| \underset{(M-1)\times k}{U_{s2}} - \underset{(M-1)\times k}{U_{s1}} \cdot \underset{k\times k}{\Psi} \|_F^2$

$U_{s2} = U_{s1}\Psi$ 唯一解或无解 $\quad \boxed{\square}\,\square = \boxed{\square}$

求 Ψ 特征值：$\lambda_1 \cdots \lambda_k$，$\quad \lambda_k = e^{j\pi\cos\theta_k}$

即可求相位 $\pi\cos\theta_k$

原理：取 $A_1 := A$ 前 $M-1$ 行 $\quad A_2 := A$ 后 $M-1$ 行

因为 $A(P + \frac{1}{M}\sigma^2 I)A^H = R = \underset{M\times k}{U_s}\underset{k}{\Lambda_s}U_s^H + \underset{(M-k)}{U_n}\Lambda_n U_n^H$

有 $\underset{M\times k}{U_s} = \underset{M\times k}{A} \cdot \underset{k\times k}{T}$ $\quad$ 也称 $\text{span}(A) = \text{span}(U_s)$

则纵向切片 $U_{s1} = A_1 T$，$U_{s2} = A_2 T$

$$\text{ULA} \longrightarrow \underset{(M-1)\times k}{A_2} = \underset{(M-1)\times k}{A_1} \cdot \begin{bmatrix} e^{j\pi\cos\theta_1} & & \\ & \ddots & \\ & & e^{j\pi\cos\theta_k} \end{bmatrix}$$

$$\begin{matrix} & \boxed{\begin{matrix}1\\ e\\ e^2\\ e^3\end{matrix}} \\ A_1 & \\ A_2 & \end{matrix}$$

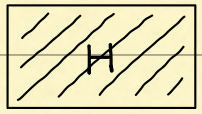
故 $U_{s2} = U_{s1}\Psi$，其中 $\Psi = T^{-1}\begin{bmatrix} e^{j\pi\cos\theta_1} & & \\ & \ddots & \\ & & e^{j\pi\cos\theta_k} \end{bmatrix} T$

$y_m = \sum_{k=1}^{K} x_k e^{jmz_k}$ 无论是 $\sum_{n=1}^{N} x_k e^{j2\pi mn/N}$ 或 $\sum_{k=1}^{K} x_k e^{j\pi m\cos\theta_k}$

$y = Ax$ 关心的是相位 z_k

ULA $\longrightarrow$ A Vandermonde

Prony Method 条件：ULA



构建 Hankel 矩阵 $\underset{(M-K)\times(K+1)}{H} = (y_{i+j})_{i=0}^{M-K-1}{}_{j=0}^{K}$

求解 $H \cdot \begin{bmatrix} p \\ 1 \end{bmatrix}_{K\times 1} = 0$

Prony polynomial $p(z) = z^K + p_{K-1}z^{K-1} + \cdots + p_1 z + p_0$

求 $p(z)=0$ 的根即为 $z_1 \cdots z_K$

缺点：条件数太大 $\mathrm{Cond}(H) := \|H^{-1}\| \cdot \|H\| = \frac{\sigma_{max}(H)}{\sigma_{min}(H)}$

Hankel 矩阵如 Hilbert 矩阵是典型病态阵

ESPIRA

求 y 的 DFT. $\hat{y} = Fy$

则可构造有理函数 必须是 ULA 才能算

$$r(z) = \sum_{k=1}^{K} \frac{a_k}{z - z_k}$$

M 个插值点 (z_m, r_m)，用 AAA 算法求 $r(z)$ 表达式

再用分解算法求 a_k, z_k 进而求 θ_k. x_k

$$\hat{y}_t = \sum_{m=1}^{M} y_m e^{j2\pi mt/M} = \sum_{m=1}^{M} e^{j2\pi mt/M} \left[\sum_{k=1}^{K} x_k e^{j2\pi mk/N} \right]$$

$$= \sum_{k=1}^{K} x_k \left[\sum_{m=1}^{M} e^{j2\pi m\left(\frac{t}{M} + \frac{k}{N}\right)} \right]$$

Let $\phi = e^{j2\pi\left(\frac{t}{M} + \frac{k}{N}\right)}$

因为 ULA，$\sum_{m=1}^{M} \phi^m = \phi \cdot \frac{1-\phi^M}{1-\phi}$

$$= \sum_{k=1}^{K} x_k \frac{1 - e^{-j2\pi \cancel{M}\left(\frac{t}{\cancel{M}} + \frac{k}{N}\right)}}{e^{-j2\pi\left(\frac{t}{M} + \frac{k}{N}\right)} - 1}$$

$$\underbrace{\boxed{\hat{y}_t e^{-j2\pi \frac{t}{M}}}}_{r_1 \cdots r_M} = \sum_{k=1}^{K} \frac{\overbrace{\boxed{x_k(1 - e^{-j2\pi Mk/N})}}^{a_k \text{ unknown}}}{\underbrace{\boxed{e^{-j2\pi \frac{k}{N}}}}_{z_k \text{ unknown}} - \underbrace{\boxed{e^{j2\pi \frac{t}{M}}}}_{z_1 \cdots z_M}}$$

Gaussian Sum :

$N \equiv 3 \pmod 4$, $M = \frac{1}{2}(N-1)$, p coprime with N

$$\sum_{m=1}^{N} e^{j2\pi p m^2/N} = \left(\frac{p}{N}\right) j\sqrt{N}$$

$$\sum_{m=1}^{M} e^{j2\pi p m^2/N} = \frac{\pm j\sqrt{N} - 1}{2}$$

$$\hat{y}_t = \sum_{m=1}^{M} y_m e^{j2\pi m^2 t/N} = \sum_{m=1}^{M} e^{j2\pi m t/M} \left[\sum_{k=1}^{K} x_k e^{j2\pi m^2 k/N} \right]$$

将未采样点设为0

做 N 维 DFT

$$= \sum_{k=1}^{K} x_k \left[\sum_{m=1}^{M} e^{j2\pi m^2 \cdot (t+k)/N} \right]$$

$$= \begin{cases} \frac{1}{2}\left[\left(\frac{t+k}{N}\right) j\sqrt{N} - 1\right], & t+k \text{ 与 } N \text{ 互质，就是 } t+k \neq N \\ M, & t+k = N \end{cases}$$

$$= M x_{N-t} + \sum_{k \neq N-t} x_k \frac{1}{2}\left[\left(\frac{t+k}{N}\right) j\sqrt{N} - 1\right]$$

Let $\alpha = \frac{1}{2}(j\sqrt{N} - 1)$, $\beta = \frac{1}{2}(-j\sqrt{N} - 1)$

$$\hat{y} = \begin{pmatrix} \alpha & \alpha & \beta & \beta & M \\ \alpha & \beta & \beta & M & \alpha \\ \beta & \beta & M & \alpha & \alpha \\ \beta & M & \alpha & \alpha & \beta \\ M & \alpha & \alpha & \beta & \beta \end{pmatrix}$$

Hankel

left shift

$x_{N\times 1}$ with sparsity K